

FTP & TFTP

FTP (File Transfer Protocol) and **TFTP (Trivial File Transfer Protocol)** are used to **transfer files over a network**.

- Both use a **Client–Server model**.
- Files can be transferred **from server → client** or **client → server**.
- Common networking use: **upgrading the IOS/software of network devices**.

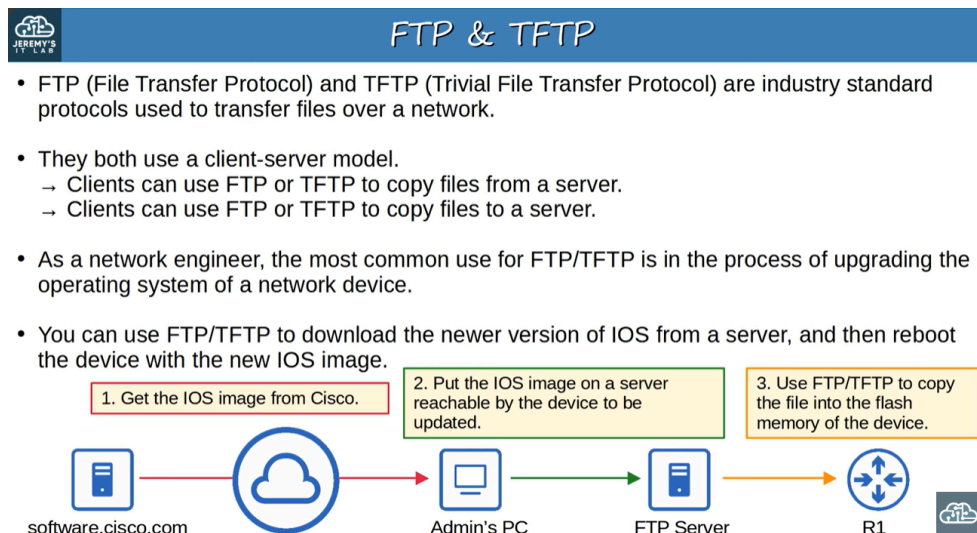
IOS Upgrade Process

1. Get the new **IOS image from Cisco**.
2. Place the IOS image on an **FTP/TFTP server**.
3. Use FTP/TFTP to copy the image into the device's **Flash memory**.
4. **Reboot** the device using the new IOS image.

Example:

Cisco → Admin PC → FTP/TFTP Server → Router R1

Exam point: FTP/TFTP are commonly used to transfer IOS images and configuration files to/from Cisco devices.



TFTP Reliability

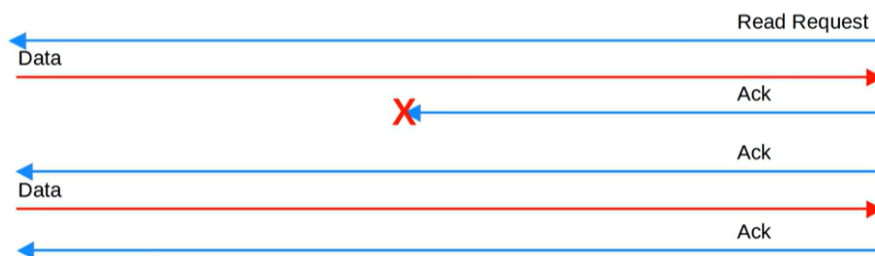
TFTP uses UDP, so UDP itself does NOT provide reliability. TFTP adds its own simple reliability mechanism using ACKs, timers, and retransmissions.

- TFTP uses **UDP** → connectionless and no built-in reliability.
- But TFTP provides **its own basic reliability**.
- **Every TFTP data block must be acknowledged (ACK).**
- Uses **lock-step communication**:
 - Send **one data block**
 - Wait for **ACK**
 - Then send the **next data block**
- Uses a **timer**.
- If ACK is not received before timeout → **retransmit the previous data block**.
- Therefore, TFTP is **more reliable than raw UDP**, but not as feature-rich/reliable as TCP.



TFTP Reliability

- Every TFTP data message is acknowledged.
 - If the client is transferring a file to the server, the server will send Ack messages.
 - If the server is transferring a file to the client, the client will send Ack messages.
- Timers are used, and if an expected message isn't received in time, the waiting device will re-send its previous message.



TFTP uses 'lock-step' communication. The client and server alternately send a message and then wait for a reply. (+retransmissions are sent as needed)



TFTP "Connections"

A TFTP file transfer has **3 phases**:

1. Connection / Initialization

- Client sends a **request (RRQ/WRQ)** to the server.
- Server responds → transfer begins.

2. Data Transfer

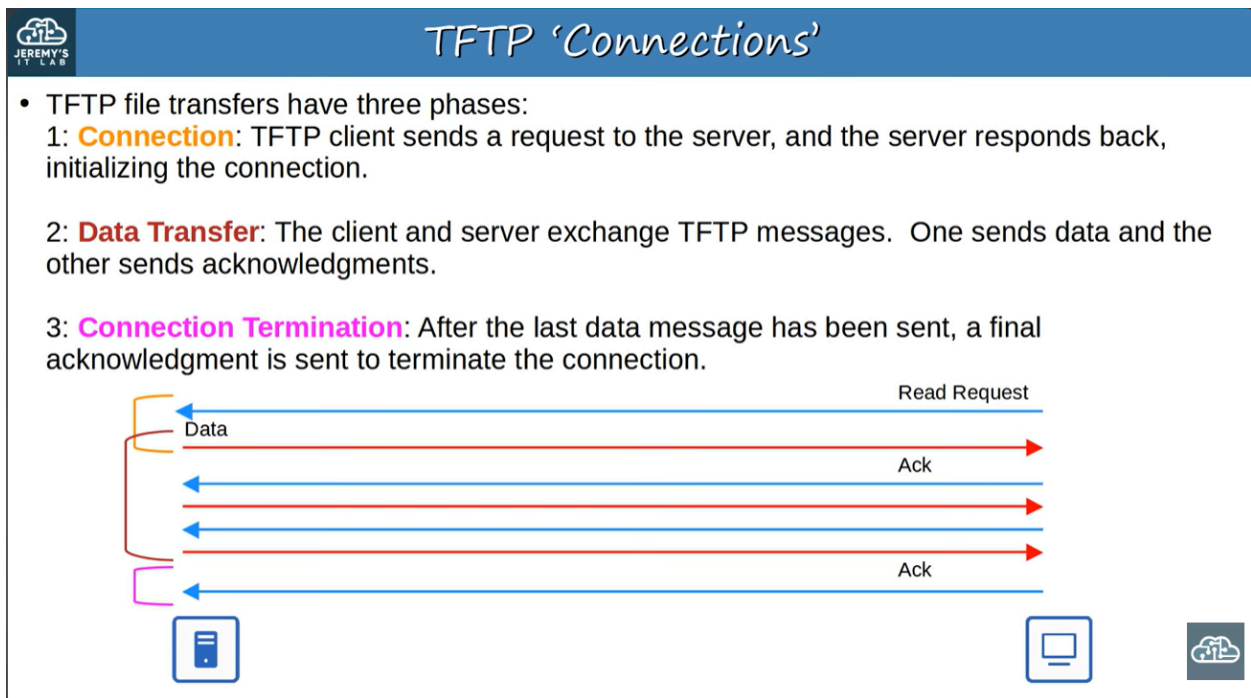
- Client and server exchange **DATA and ACK** messages.
- **One DATA → one ACK → next DATA.**
- Uses **UDP** with TFTP's own ACK/retransmission mechanism.

3. Connection Termination

- After the **last DATA block**, the receiver sends the **final ACK**.
- Transfer ends.

🔑 Flow to Remember

Request → DATA → ACK → DATA → ACK → ... → Last DATA → Final ACK → End

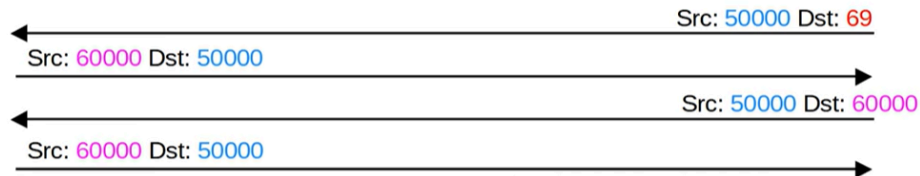


TRANSFER IDENTIFIER



TFTP TID

- When the client sends the first message to the server, the destination port is UDP 69 and the source is a random ephemeral port.
- This random port is called a 'Transfer Identifier' (TID) and identifies the data transfer.
- The server then also selects a random TID to use as the source port when it replies, **not 69**.
- When the client sends the next message, the destination port will be the server's TID, **not 69**.



FTP

- **FTP = File Transfer Protocol**
- Standardized in **1971**.
- Uses **TCP**:
 - **Port 20** → **Data**
 - **Port 21** → **Control/commands**
- Uses **username + password** for authentication.
- Traditional FTP has **no encryption** → credentials/data can be exposed.
- **FTPS** → FTP secured with **SSL/TLS**.
- **SFTP** → Secure File Transfer Protocol over **SSH**.
- More feature-rich than TFTP:
 - Transfer files
 - Browse/list directories
 - Create/delete directories
 - Other file-management operations
- Client sends **FTP commands** to the server.

Remember

FTP → TCP → 20 (Data), 21 (Control)

FTPS → FTP + SSL/TLS

SFTP → SSH-based secure file transfer

FTP Connections

FTP uses **2 separate TCP connections** because it separates **control** from **actual file/data transfer**.

1. FTP Control Connection

- Uses **TCP port 21**
- Used for **commands and replies**
- Remains open during the FTP session.
- Example commands: login, request file, list directory, etc.

Client — Commands —→ Server

Client ←— Replies — Server

2. FTP Data Connection

- Used for **actual file/data transfer**.
- In **Active Mode**, the **server initiates** the data connection.
- Traditionally uses **TCP port 20** on the server.
- Created/terminated as needed for data transfers.

Client ←—— File/Data ———→ Server

Why are there TWO connections?

Think of it like this:

CONTROL CONNECTION

"Hey server, give me file.txt."

"Okay, I'll send it."

DATA CONNECTION

"Here is file.txt."

So:

Port 21 = What should we do?

Port 20 = Here is the actual data.

This separation allows FTP to send **commands/control information** independently from the **large file data**.

 **Active Mode**

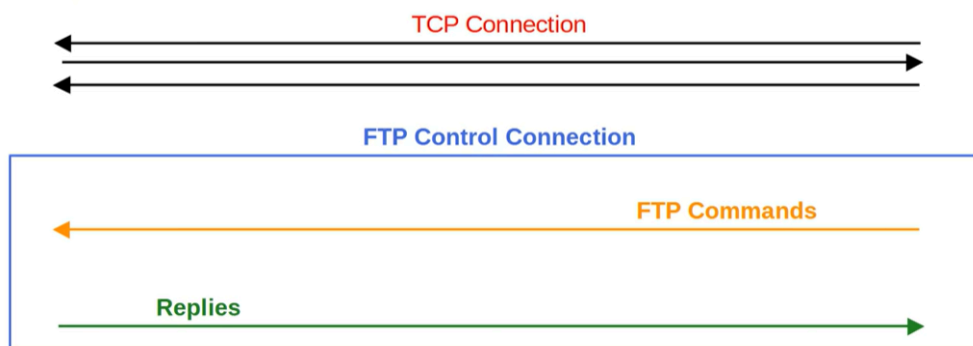
In **Active FTP**:

Client establishes the control connection → Server initiates the data connection.

 **Image placement:**

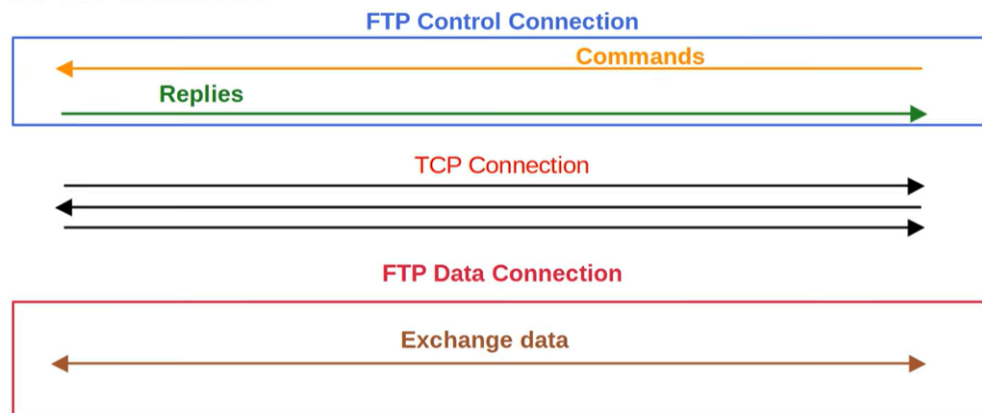
FTP Control Connections

- FTP uses two types of connections:
 - An **FTP control** connection (**TCP 21**) is established and used to send FTP commands and replies.
 - When files or data are to be transferred, separate **FTP data** (**TCP 20**) connections are established and terminated as needed.



Active Mode FTP Data Connections

- The default method of establishing FTP data connections is **active mode**, in which the server initiates the TCP connection.



In FTP **active mode**, the server initiates the data connection.



One important clarification

Don't think there are **two active connections doing the same thing**.

There are **two different TCP connections with different jobs**:

Connection	Purpose	Traditional Port
Control	Commands + replies	TCP 21
Data	File/data transfer	TCP 20 (active mode)

Memory trick:

🧠 **21 = Control** → "What do you want?"

🧠 **20 = Data** → "Here is the file."

FTP Passive Mode

Passive Mode

- In **Passive FTP**, the **client initiates both connections**:
 - Control connection
 - Data connection
- Useful when the **client is behind a firewall**.
- The firewall may block an **incoming connection initiated by the server**.
- Server tells the client which **dynamic port** to use for the data connection.
- Client then connects to that server port.

Active vs Passive

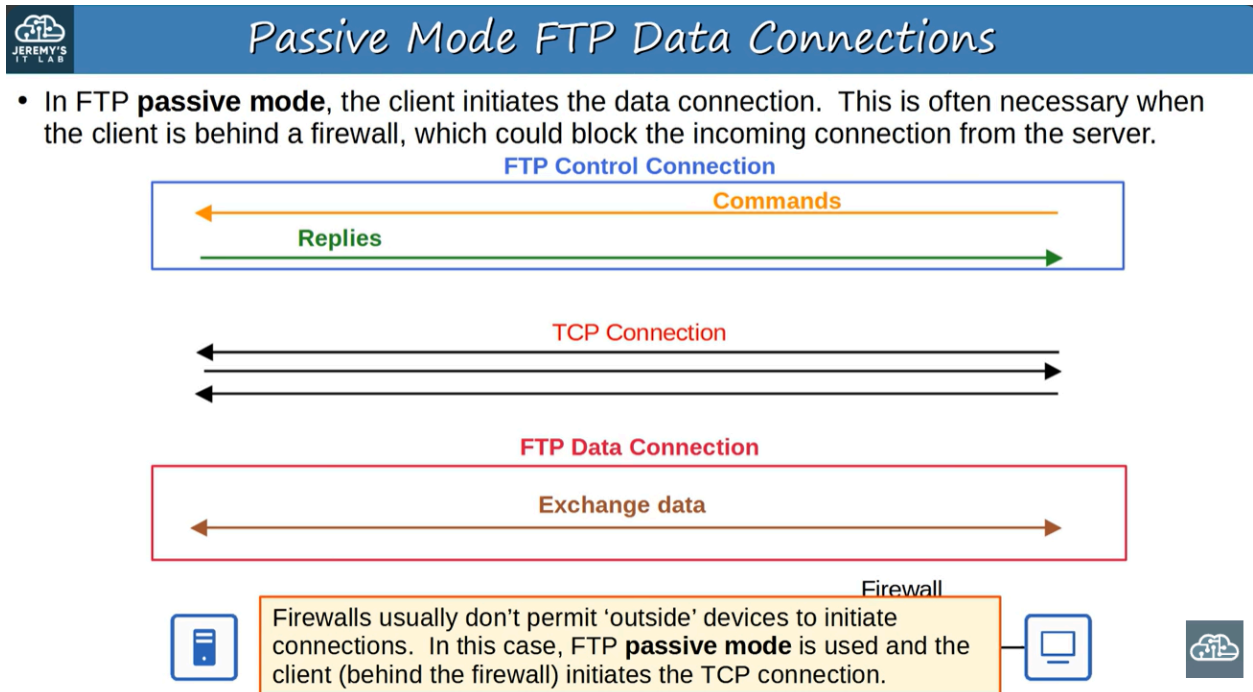
Mode	Who initiates Data Connection?	Common use
Active	Server	Client allows incoming connections
Passive	Client	Client behind firewall/NAT

🔑 Easy Memory

Active → Server initiates data

Passive → Client initiates data

The **control connection** is still **TCP port 21** in both modes



FTP

- Uses TCP (20 for data, 21 for control) for connection-based communication
- Clients can use FTP commands to perform various actions, not just copy files
- Username/PW authentication
- More complex

TFTP

- Uses UDP (69) for connectionless communication (although a basic form of 'connection' is used within the protocol itself)
- Clients can only copy files to or from the server
- No authentication
- Simpler

IOS File Systems

A **file system** controls how data is **stored and retrieved**.

View IOS File Systems

show file systems

Important File Systems

Type	Meaning
disk:	Storage devices such as Flash memory
nvr:	Internal NVRAM; stores startup-config
network:	External file systems such as FTP/TFTP servers
opaque:	Used for internal IOS functions

Important Prefixes

flash: → Flash memory
 nvram: → Startup configuration
 tftp: → TFTP server
 ftp: → FTP server



IOS File Systems

- A file system is a way of controlling how data is stored and retrieved.
- You can view the file systems of a Cisco IOS device with **show file systems**

```
Router#show file systems
File Systems:
```

Size(b)	Free(b)	Type	Flags	Prefixes
* 2142715904	1994403840	disk	rw	flash0: flash:#
-	-	disk	rw	flash1:
966656	962560	disk	rw	flash2:#
-	-	disk	rw	flash3:
-	-	opaque	rw	archive:
-	-	opaque	rw	system:
262144	256791	nvr	rw	nvr
-	-	opaque	rw	tmpsys:
-	-	network	rw	snmp:
-	-	opaque	rw	null:
-	-	network	rw	tftp:
-	-	opaque	ro	xmodem:
-	-	opaque	ro	ymodem:
-	-	opaque	wo	syslog:
-	-	network	rw	rcp:
-	-	network	rw	pram:
-	-	network	rw	ftp:

[output omitted]

disk: Storage devices such as flash memory.

opaque: Used for internal functions

nvr: Internal NVRAM. The startup-config file is stored here.

network: Represents external file systems, for example external FTP/TFTP servers.

Cisco IOS Upgrade + FTP/TFTP

1. Check Current IOS

show version

→ Displays the **current IOS version**.

show flash

→ Displays IOS files stored in **Flash**.



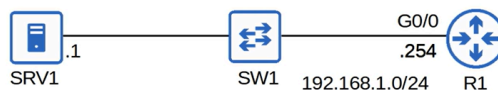
Upgrading Cisco IOS

- You can view the current version of IOS with **show version**

```
R1#show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.1(4)M4, RELEASE SOFTWARE
(fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2012 by Cisco Systems, Inc.
Compiled Thurs 5-Jan-12 15:41 by pt_team
[output omitted]
```

- You can view the contents of flash with **show flash**

```
R1#show flash
System flash directory:
File Length Name/status
3 33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
2 28282 sigdef-category.xml
1 227537 sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```



2. Copy IOS from TFTP Server to Flash

General syntax:

copy	source	destination
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9
10	10	10
11	11	11
12	12	12
13	13	13
14	14	14
15	15	15
16	16	16
17	17	17
18	18	18
19	19	19
20	20	20
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92	92	92
93	93	93
94	94	94
95	95	95
96	96	96
97	97	97
98	98	98
99	99	99
100	100	100

Example:

```
copy tftp: flash:
```

Then enter:

1. **TFTP server IP**
2. **Source filename** on server
3. **Destination filename** on Flash

[illegible]

3. Configure Router to Boot New IOS

conf t

boot system flash:<IOS-filename>

exit

write memory

reload

- **boot system** → tells router **which IOS to boot**.
- **write memory** → saves the boot setting.
- **reload** → restarts router.



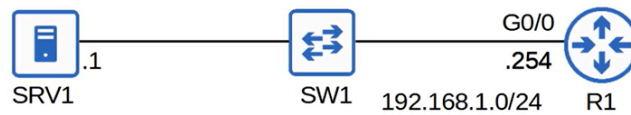
Upgrading Cisco IOS

```
R1#show flash

System flash directory:
File Length Name/status
 3 33591768 c2900-universalk9-mz.SPA.151-4.M4.bin
 4 33591768 c2900-universalk9-mz.SPA.155-3.M4a.bin
 2 28282 sigdef-category.xml
 1 227537 sigdef-default.xml
[67439355 bytes used, 188304645 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```

boot system filepath
*If you don't use this command, the router will use the first IOS file it finds in flash

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#boot system flash:c2900-universalk9-mz.SPA.155-3.M4a.bin
R1(config)#exit
R1#write memory
Building configuration...
[OK]
R1#reload
Proceed with reload? [confirm]
```



If **boot system** is not configured, the router may use the **first IOS file it finds in Flash**.

4. Remove Old IOS

delete flash:<old-IOS-filename>

Then verify:

show flash

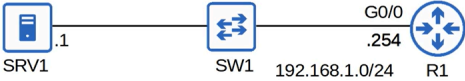
Upgrading Cisco IOS

```
R1#show version
Cisco IOS Software, C2900 Software (C2900-UNIVERSALK9-M), Version 15.5(3)M4a, RELEASE SOFTWARE(fc1)
[output omitted]

R1#delete flash:c2900-universalk9-mz.SPA.151-4.M4.bin
Delete filename [c2900-universalk9-mz.SPA.151-4.M4.bin]?
Delete flash:/c2900-universalk9-mz.SPA.151-4.M4.bin? [confirm]

R1#show flash

System flash directory:
File Length Name/status
 4 33591768 c2900-universalk9-mz.SPA.155-3.M4a.bin
 2 28282 sigdef-category.xml
 1 227537 sigdef-default.xml
[33847587 bytes used, 221896413 available, 255744000 total]
249856K bytes of processor board System flash (Read/Write)
```



FTP File Transfer

Before using FTP, configure the FTP credentials:

```
conf t
```

```
ip ftp username <username>
```

```
ip ftp password <password>
```

Then:

```
copy ftp: flash:
```

Enter:

1. **FTP server IP**
2. **Source filename**
3. **Destination filename**



Copying Files (FTP)

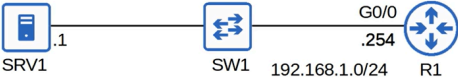
```

R1(config)#ip ftp username cisco
R1(config)#ip ftp password cisco
R1(config)#exit

R1#copy ftp: flash:
Address or name of remote host []? 192.168.1.1
Source filename []? c2900-universalk9-mz.SPA.155-3.M4a.bin
Destination filename [c2900-universalk9-mz.SPA.155-3.M4a.bin]?

Accessing ftp://192.168.1.1/c2900-universalk9-mz.SPA.155-3.M4a.bin...
Loading c2900-universalk9-mz.SPA.155-3.M4a.bin from
192.168.1.1: !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
[output omitted]
```

Configure the FTP username/password that the device will use when connecting to an FTP server.



Command Revision

show file systems

show version

show flash

copy source destination

boot system flash:<filename>

ip ftp username <username>

ip ftp password <password>

Easy Flow

Check IOS



show version

↓

Check Flash

↓

show flash

↓

Copy new IOS

↓

copy tftp: flash:

OR

copy ftp: flash:

↓

Set boot IOS

↓

boot system flash:<filename>

↓

Save

↓

write memory

↓

Reload

↓

Delete old IOS if needed



Command Review

```
R1# show file systems
R1# show version
R1# show flash
R1# copy source destination
R1(config)# boot system filepath
R1(config)# ip ftp username username
R1(config)# ip ftp password password
```